

Cloud-Based Auditing System with Efficient Ownership Management for Group Data Transactions

A complete working implementation of an IEEE 2025 research protocol, with measured proof and executed attacks.
Dept. of Information Science & Engineering, Global Academy of Technology · Major Project Phase I (ISEP23605), AY 2026-27 · Base paper: Wu, You & Huang, *IEEE Transactions on Information Forensics and Security*, vol. 20, 2025 · DOI 10.1109/TIFS.2025.3636056

1. The problem

You put a very large dataset in the cloud. Two awkward questions follow, and ordinary software cannot answer either one.

Is my data still there, and still correct?

You cannot download 100 GB every week to check. The cloud provider has a quiet financial incentive to delete data nobody reads, and then claim it is fine. Disks also fail on their own, silently.

What happens when I sell it?

Datasets are traded every day: hospitals, banks, labs, AI vendors. The buyer must genuinely gain the power to check the data, and the seller must genuinely lose it. Editing a row in a database does not achieve that.

2. Why existing research is not enough

Researchers solved the first question decades ago with **Provable Data Possession (PDP)**: a way to prove a file is intact without downloading it. It was later extended to allow selling, called **DT-PDP**. Both work well for a **single owner**. They break down when data is owned by a **group**, which is how valuable data is actually held: three hospitals, two sequencing labs, or a bank plus its custodian. Two specific failures follow.

THE FAILURE	WHAT IT COSTS IN PRACTICE
No scalability. A group is treated as one fixed block. Adding or removing a single member is handled as a full group-to-group handover.	Cost grows with the square of the number of owners, written $O(s^2)$. A 20-member consortium costs 400 units of work, not 20.
A secret leaks during the sale. The protocol requires the seller to hand its index secret key to the buyer in plain sight.	A buyer who then teams up with the cloud, a collusion attack , can forge proofs for data that was never sold . The seller is told their data is safe when it is already gone.

3. What this project delivers

A complete implementation of the protocol that fixes both, and, crucially, **proof by execution**: real attacks run against real code, with performance measured rather than claimed.

10.5x Faster group setup at 20 owners, and the advantage <i>grows</i> with group size	174x Faster to remove an owner from a 40-member group	160 bytes To transfer ownership, of <i>any</i> dataset size, <i>any</i> group size
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In one sentence: a group of hospitals can prove their dataset is intact in the cloud **without downloading any of it**, add and remove members freely, and then **sell it**, after which the buyer can verify the data and the sellers provably cannot.

4. Verified status

COMPONENT	EVIDENCE	STATUS
All 6 protocol algorithms	41 automated tests	41 / 41 PASSING
Full lifecycle demonstration	Upload → audit → add → remove → sell	7 / 7 AS EXPECTED
Large sample datasets	5 real-world sale scenarios, up to 512 MB	GENERATED
Three security attacks	Each run against old method <i>and</i> ours	ALL DEFENDED
Performance benchmarks	39-minute measured run, 6 result charts	MEASURED

5. The four parties

The system models four roles. What matters most is **who is trusted with what**: the design assumes the cloud may lie.

PARTY	WHO THEY ARE	TRUSTED?	NEVER SEES
Sale Group (SG)	The owners today: e.g. 3 hospitals	Yes, but no one acts alone	Each other's secret keys
Cloud (CSP)	The storage company	No , assumed it may delete or lie	Any secret key or mask
Auditor (TPA)	Independent inspector	Yes, to stay impartial	The data itself, ever
Buy Group (BG)	The purchaser	Must <i>earn</i> ownership	Any seller secret key

6. The six algorithms, in plain words

- 1 SysGen, set up the playing field**
Creates the shared mathematical parameters. Critically, two special numbers are generated so that **nobody, not even us**: can know the hidden relationship between them. If anyone could, they could forge every proof in the system.
- 2 KeyGen, every owner makes its own keys**
Each owner independently creates a private and public key: no meeting, no shared secret. **This is the step that is 10× faster**, because each owner is bound using a fingerprint of its *own* key rather than of everyone else's.
- 3 TagGen, seal the data**
The file is cut into blocks. For each block every owner creates a cryptographic "wax seal", a **homomorphic tag** built with **BLS signatures**. The seals multiply into **one seal per block** containing all owners. No owner can remove another's contribution; no outsider can fake one.
- 4 Audit, the inspection**
The auditor picks blocks **at random** and demands proof. The cloud replies with just **two values (~128 bytes)**: never the data. Randomness is essential: a predictable check would let the cloud keep only those blocks and bin the rest.
- 5 OwnerModify, members join and leave**
A joining or leaving owner sends one small **Ownership Dynamic Token (ODT)**; the cloud updates every seal. Existing owners do nothing: no re-keying, no need to even be online.
- 6 OwnerTransfer, the sale**
Sellers cancel their contribution and produce a token; buyers add theirs and fold the two into **one** token sent to the cloud. The buyer never receives any seller secret: only a blinded, aggregated token.

7. The single trick that makes it secure

Every owner mixes a secret **random mask** into every seal. Only a *combined total* is ever published, and a total cannot be reversed back into its parts.

When an owner leaves, their mask is **deliberately left behind**, baked into every seal forever. It looks like a bug. It is the entire defence: even if a buyer obtains the key the old system leaked, the sellers' masks are still welded in and cannot be stripped out. *We wrote a test that fails loudly if anyone ever "cleans this up."*

8. What the live demo shows

Running `python3 -m demo.run_demo` plays out a full story on a real 25 MB hospital dataset, and the auditor's verdict is checked at every stage:

STEP	WHAT HAPPENS	AUDITOR RESULT
1–4	3 hospitals set up keys and seal 33 blocks of real data	—
5	Honest cloud is inspected	PASS ✓
6	Cloud secretly corrupts a block	FAIL ✓ caught, this failure is a success
7	A fourth hospital joins the consortium	PASS ✓
8	An original hospital leaves	PASS ✓
9a	Dataset is SOLD , buyer inspects it	PASS ✓
9b	Seller tries with their old credentials	FAIL ✓ ownership truly moved

Step 9b is the most important line in the project. Ownership did not change in a database row someone could edit back, it changed *mathematically*. Neither the seller nor a corrupt employee can undo it.

9. The three attacks, the strongest part of this project

Anyone can claim a system is secure. Each attack is **executed twice**: once against the older published method, once against ours. Showing only the defence would prove nothing: a system that rejects everything also rejects attacks.

ATTACK	OLDER PUBLISHED METHOD	THIS PROJECT
Collusion Buyer teams up with the cloud, using the key the old protocol forces the seller to hand over	BROKEN, accepted a proof for a block the cloud had deleted	BLOCKED, identical attack rejected
Rogue-key attack Faking co-ownership of a dataset belonging to strangers	BROKEN, the forgery was accepted as genuine	BLOCKED, and real signatures still work
Data tampering Malice, or an ordinary failing disk	CAUGHT EVERY TIME, including a block deleted and replaced with zeros	

In the collusion test the attacker was deliberately handed **more secret information than the real protocol would ever leak**. The attack still failed. That is a far stronger result than merely following the rules.

10. Measured performance

The paper's central claim is that cost should grow *in step with* group size, **O(s)**, rather than *with its square*, **O(s²)**. Tested directly by doubling the group:

GROUP SETUP COST	10 OWNERS	20 OWNERS	GROWTH ON DOUBLING
This project	746 ms	1,486 ms	1.99× (ideal 2×)
Older method	4,076 ms	15,540 ms	3.81× (ideal 4×)

Doubling the group **doubles** our cost but nearly **quadruples** theirs, so the advantage widens as groups grow: **1.1×** at one owner, **10.5×** at twenty, roughly **50×** projected at a hundred.

OPERATION	THIS PROJECT	OLDER METHOD	ADVANTAGE
Add 20 owners to a group of 20	2,445 ms	60,568 ms	25×
Remove 5 owners from a group of 40	272 ms	47,376 ms	174×
Transfer ownership (20 ↔ 20 owners)	3,377 ms	—	160-byte token

Checking is free of group size

An audit took **692.6 ms** with 1 owner and **692.1 ms** with 10, a **0.07 %** difference. The auditor genuinely cannot tell how many owners exist, so consortia can grow without slowing anyone down.

Checking is free of data size

Every audit costs the same **three cryptographic pairings** whether the dataset is 1 KB or 100 TB, and the cloud's reply is about **128 bytes**. The data itself never moves.

11. An original finding

The paper estimates how likely an audit is to catch a cheating cloud, using a standard formula. We measured it over **3,500 trials** and found the real detection rate **meets or beats the paper's figure in every configuration tested**.

The reason: their formula assumes the auditor might check the same block twice; ours never does. Their recommendation is therefore *safe*, it **under-promises** what the system actually delivers. This is a small original contribution back to the research, and a strong point to raise with an examiner.

12. Engineering quality

41

Automated tests, all passing

6,058

Lines of documented Python

3

External dependencies only

500 MB

Sample data generated

13. What has been delivered

FOLDER	CONTENTS
crypto/	The protocol itself: all six algorithms, no file/network/database code mixed in. Readable on its own.
entities/	The four parties as plain classes: Cloud, Auditor, Sale Group, Buy Group.
datasets/	Generator for five realistic sale scenarios at four size tiers (2 MB → 512 MB).
demo/	The narrated end-to-end walkthrough judges can watch live.
attacks/	Three executable attack demonstrations.
benchmarks/	Measurement harness plus six publication-quality result charts.
tests/	41 correctness tests.
report/	Chapters 5–7 (Implementation, Testing, Results) and all Chapter 4 diagrams.

14. The five built-in market scenarios

Each dataset is deliberately owned by a **consortium**, never one party, because that is the case existing research fails on.

DATASET	SELLER	BUYER
Medical imaging index	3 hospital groups	Medical-AI research institute
Smart-city telemetry	Transport dept + 2 agencies	Urban analytics company
Banking ledger	Bank + data custodian	Statutory audit firm
Genomic variants	3 sequencing labs	Pharmaceutical company
AI training corpus	Data-labelling vendor	Autonomous-systems AI lab

All data is synthetic: no real patient, customer or personal record is used or reproduced.

15. Honest limitations

- **Speed is in milliseconds, not microseconds.** We chose a pure-Python cryptography library so the project runs on *any* machine with no compiler; the original authors used C. This affects raw speed only, *not* the scaling behaviour being demonstrated, since both methods are measured on identical foundations. All cryptography sits in one file and can be swapped for a native version.
- **The comparison system is a faithful re-build** of the older method's cost structure as described in the paper, not the original authors' code. Its shape is accurate; its exact millisecond values are ours.
- **No graphical interface yet.** This phase delivers the engine, datasets, security proofs and measurements.
- **Not production-hardened.** A real deployment needs secure key storage, access control and formal review.

16. Future goals

PHASE	GOAL	WHY IT MATTERS
Next	Native cryptography backend	100x speed gain. One file changes; nothing else moves.
Next	Web dashboard for the four roles	Makes the system usable by non-programmers and demo-ready.
Phase II	Real cloud deployment (AWS S3 / Azure)	Proves it works against a genuine storage provider, not a simulation.
Phase II	Automatic scheduled auditing + alerts	Continuous protection instead of manual checks.
Research	Partial-dataset sale	Sell half a dataset and keep the rest: a real commercial need.
Research	Blockchain ownership history & post-quantum variant	A tamper-proof public record of who owned what; and resistance to future quantum computers.
Publication	Paper on the detection-bound finding	An original, defensible contribution already measured.

17. Why this matters commercially

The scenarios above are real markets: **hospital consortia** selling imaging data to medical-AI companies, **banks** handing ledgers to statutory auditors, **sequencing labs** licensing genomic data to pharma, **data vendors** selling AI training sets worth millions. All share one unsolved need (proving a dataset is intact and that its ownership genuinely moved) and **none has a single owner**. That is the gap this project closes.